

Practice 1

Exercise levels:

(L1) *Reproduce*: Reproduce basic facts or check basic understanding.

(L2) *Apply*: Follow step-by-step instructions.

(L3) *Reason*: Show insight using a combination of different concepts.

(L4) *Create*: Prove a non-trivial statement or create an algorithm or data structure of which the objective is formally stated.

► Lecture 0 Prior Knowledge

Exercise 1 (L2) Use the logarithmic identities (e.g., to be found at wikipedia) to simplify the following expressions as much as possible.

(a) $\log_2(128) + \log_{10}(100)$

(b) $\log_2(64) - \log_3(27)$

(c) $\log_2\left(\frac{1}{4}\right) + \log_3(9^4)$

(d) $\log_2(16n^2) - \log_2(2n^4) + \log_2\left(\frac{32}{\sqrt{n}}\right)$

Exercise 2 (L2) Compute or simplify the following sums. If needed, check Appendix A for helpful formulas and properties.

(a) $\sum_{i=0}^3 (i^2 + 2i + 4)$

(b) $\sum_{j=1}^2 (-1)^j \frac{n^j}{j^2}$

Question for the interested reader: What happens if we start the summation with $j = 0$?

(c) $\sum_{k=3}^6 \frac{2^{k+1}x^k}{k}$

(d) $\sum_{j=2}^4 \sum_{i=0}^4 (i + j)$

(e) $\sum_{k=0}^n (k^2 - (k + 1)^2)$

Exercise 3 (L1) Rewrite the given sums as summations.

(a) $2 + 5 + 8 + 11 + 14 + 17$

(b) $x^7 - \frac{x^9}{1!} + \frac{x^{11}}{2!} - \frac{x^{13}}{3!} + \frac{x^{15}}{4!} - \frac{x^{17}}{5!}$

► Lecture 1 Introduction

Exercise 4 (L1) Prof. C. Monster has an array A which he needs to sort. He knows that A contains elements with the keys $\{1, 1, 1, 2, 2, 4\}$, but he does not know in which order. Prof. C. Monster reckons that, because he knows exactly which values are stored in the array, he can simply overwrite them with the values in the right order. Explain why satellite data are sometimes important, and why his approach would lead to problems.

Exercise 5 (L2) (This is (part of) Exercise 2.1-4 from the book.) Consider the searching problem:

Input: A sequence of n numbers $\langle a_1, a_2, \dots, a_n \rangle$ stored in array $A[1 : n]$ and a value x .

Output: An index i such that x equals $A[i]$ or the special value NIL if x does not appear in A .

- Give a natural language description of how an algorithm could solve this problem.
- Write pseudocode for linear search, which scans through the array from the beginning to the end, looking for x .

Exercise 6

- (L1) Give definitions of the big- O , big- Ω , and big- Θ notations.
- (L3) Explain using the definitions why $f(n) = \Theta(n)$ when $f(n) = \Omega(n)$ and $f(n) = O(n)$.

Exercise 7 (L2) For each function on the left, $p(n)$, write the letter of a function on the right, $q(n)$, such that $p(n) \in \Theta(q(n))$. If no such function $q(n)$ is listed, then choose (l).

$f(n) = \sum_{i=1}^n (4i - 4)$	_____	(a) 1	(g) $\log n$
$g(n) = \sum_{i=1}^n \sum_{j=1}^i i$	_____	(b) n	(h) $n \log n$
$h(n) = \sum_{i=1}^{\log n} n$	_____	(c) $n(\log n)^2$	(i) n^2
$k(n) = \sum_{i=0}^n \frac{4}{2^i}$	_____	(d) $n^2 \log n$	(j) n^3
		(e) 2^n	(k) 2^{2n}
		(f) n^n	(l) no match

Exercise 8 (L2) Rank the following functions of n by order of growth (starting with the slowest growing). Functions with the same order of growth should be ranked equal.

$n^{\sqrt[4]{n}}$	$n^{-\log 4}$	$n^{0.00001}$	$\sum_{i=0}^n 2^i$	$4^{\log 8}$	$n!$
1	$\frac{\log(2n)}{2}$	$n \log n$	$\log \sqrt{n}$	$\sum_{i=1}^{\log n} i$	n^n

► Lecture 2 Running time analysis

Exercise 9 (L3) Formally prove or disprove the following statements using the definitions of the respective asymptotic notations:

- (a) $n^2 - 3n + 4 = \Theta(n^2)$
- (b) $n^2 \log n = \Omega(n^3)$
- (c) $\sqrt{n} = O(\log n)$
- (d) $\Omega(n \log n) \cap O(n^2) \neq \emptyset$
- (e) If $f(n) = \Omega(h(n))$ and $g(n) = \Omega(h(n))$, then $f(n)g(n) = \Omega(h^2(n))$.

Exercise 10 Consider the following algorithm, which takes an array A as input. Note that A stores n integer values.

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EVENODD(A)
1  out = 0
2  num = 0
3  for i = 1 to n
4      if A[i] is even
5          out = out + A[i]
6      else
7          num = num + 1
8  return out, num

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- (a) Give a textual description of what the algorithm does.
- (b) Formally analyze the running time of the given algorithm.

Exercise 11 (L2) Using the Master theorem, prove asymptotic upper and lower bounds for $T(n)$ in each of the following recurrences. Assume that $T(n)$ is constant for $n \leq 2$.

- (a) $T(n) = 16 T(n/4) + n^2$
- (b) $T(n) = 7 T(n/2) + n^2$
- (c) $T(n) = 4 T(n/2) + n^2 \sqrt{n}$
- (d) (L4) For the solutions of the recurrences (a)–(c), give another recurrence that solves to an exactly the same asymptotic bound, but using a different case of the Master theorem.

Exercise 12

- (a) (L3) Prove using the substitution method that the solution of $T(n) = T(\lceil n/3 \rceil) + T(\lfloor 2n/3 \rfloor) + n$ with $T(1) = 1$ is $T(n) = O(n \log n)$.
- (b) (L2) Can the master method be applied to $T(n) = 4 T(n/2) + n^2 \log \log n$? Why or why not?
- (c) (L3) Use a recursion tree to get an intuition of what the right upper asymptotic bound for $T(n) = 4 T(n/2) + n^2 \log \log n$ might be.
- (d) (L3) Prove the upper asymptotic bound from (c) using the substitution method.

► Lecture 3 Proving techniques

Exercise 13 (L2) Prove the following claims by direct proof.

- (a) We claim that $n + (n + 1) + (n + 2)$ is divisible by 3 for any $n \geq 1$.
- (b) We claim that for any even integer n its square n^2 is also even.

Exercise 14 (L3) Prove the following statement (known as the *pigeonhole principle*) using a proof by contradiction.

If n items are put into m containers and $n > m$, then there must exist a container that contains more than one item.

Exercise 15 (L2) Prove by case distinction that for every integer x it holds that $x(x + 1)$ is even.

Exercise 16 (L3) What is wrong with the following proof?

Claim: All flower bouquets have only one color of flowers.

Proof: We prove this by induction on the number n of flowers in the bouquet.

Base case: ($n = 1$)

In this case the bouquet consists of only one flower, and hence all flowers in the bouquet trivially have the same color. Thus the claim holds for $n = 1$.

Inductive step: Let $n \geq 1$.

Induction hypothesis: We assume that all flower bouquets of size n have only one color of flowers. We show the claim for a bouquet of $n + 1$ flowers:

Let B be a flower bouquet with $n + 1$ flowers $f_1, f_2, \dots, f_{n+1}$. Consider the flower bouquet B_1 formed by flowers $f_1, f_2, \dots, f_n$ and the flower bouquet B_2 formed by $f_2, \dots, f_{n+1}$. Both B_1 and B_2 consist of n flowers, and hence by the inductive hypothesis all flowers in B_1 have the same color, and likewise for B_2 . Moreover, the flowers $f_2, \dots, f_n$ are contained in both bouquets, and hence all flowers in B have the same color. This completes the inductive step.

By induction, it follows that the claim holds for flower bouquets of all sizes.

Exercise 17 (L2) Prove by induction on $n \geq 1$ that $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$.